# Weak virial positivity at every fixed order

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**Abstract**

For every fixed degree $r \geq 1$, index $i \geq 0$, and order $k \geq 2$, a uniform random simple $r$-regular bipartite graph on two labelled parts of size $n$ satisfies $\Delta^k[-\log(j!m_j)]_{j=i} > 0$ with probability $1 - O(n^{-2})$. This proves the full weak virial-positivity conjecture. The argument combines the exact Wanless–Pernici matching expansion and bounded contributor moments with an all-order formal coefficient calculation, whose proof is included.

## Task 1. Full solution

### Statement, model, and established inputs

Let $m_j = m_j(G)$ be the number of $j$-matchings and let $\Delta v(j) = v(j+1) - v(j)$. All parameters $r, i, k$ below are fixed as $n \to \infty$. The random graph is uniform among simple $r$-regular bipartite graphs with a fixed labelled bipartition of sizes $n, n$, the standard model underlying the source's random regular graph estimates. Write $u(j) = -\log(j!m_j)$.

**Theorem 1.** *For every $r \geq 1$, $i \geq 0$, and $k \geq 2$, there is a constant $C_{r,i,k}$ such that, for all sufficiently large $n$,*

$$\Pr\{\Delta^k u(i) \leq 0\} \leq C_{r,i,k} n^{-2}.$$

*In particular, Conjecture I.2 of [1] holds without the restrictions in that paper's Theorem I.1.*

We use the following two established facts in the precise form recorded in [2, Section 3, equations (3.1)–(3.10)]. Set

$$T_r(y) = \frac{r\sqrt{1-4(r-1)y} - r + 2}{2(1-r^2y)} = \sum_{s \geq 0} t_s y^s, \tag{1}$$

$$f(x) = -\sum_{s \geq 1} \frac{t_s}{s}(-x)^s, \qquad M_j(n) = [x^j]e^{nf(x)}. \tag{2}$$

Here $t_0 = 1$, $t_1 = r$, and the square root has constant term one. The rationalized form defines $r = 1$ as well.

First, the exact Wanless–Pernici expansion [3, 4] is

$$m_j = \sum_{z=0}^{j} c_z(G) M_{j-z}(n), \qquad \sum_{z \geq 0} c_z x^z = \exp\left(\sum_{s \geq 4} \frac{\eta_s(G)}{2s}(-x)^s\right). \tag{3}$$

In particular, $c_0 = 1$, $c_1 = c_2 = c_3 = 0$. Each $\eta_s$ is a fixed linear combination, with coefficients depending on $r$, of counts of finitely many fixed cyclic contributor subgraphs. Second, for every fixed finite collection of these variables and every fixed polynomial $P$,

$$\sup_{n \text{ sufficiently large}} \mathbb{E}|P(\eta_4, \dots, \eta_J)|^2 < \infty. \tag{4}$$

This is the bounded-moment input of [2, equation (3.10)]; its use is independent of that paper's finite-order coefficient restrictions.

One can see the moment bound directly in the bipartite pairing model. The expected number of embeddings of a fixed graph $H$ with $v$ vertices and $e$ edges is $O(n^{v-e})$: there are $O(n^v)$ vertex choices, boundedly many stub choices, and any $e$ prescribed pairs have probability $1/(rn)_{\underline{e}}$. In a product of cyclic contributor counts, classify the finitely many possible overlaps. Every component of each union contains a cycle, so $e \geq v$. All fixed moments are consequently bounded. Conditioning on simplicity preserves these bounds, since its probability tends to a positive constant for fixed $r$; this standard random-regular graph fact is also among the inputs cited in [2]. Polynomial expansion and Cauchy–Schwarz give (4).

## The coefficient lemma, at all orders

The next lemma is the coefficient result established in our companion MathDB 349137 solution. We reproduce its proof, including the known logarithmic degree bound, to specify every algebraic step used here.

**Lemma 2.** *Put $\varepsilon = 1/n$ and $A_j = j!M_j/(nr)^j$, with its polynomial extension in $j$. For arbitrary formal constants $c_z$, $z \geq 4$, define*

$$F_j = A_j + \sum_{z \geq 4} \frac{c_z}{r^z} \varepsilon^z (j)_{\underline{z}} A_{j-z}.$$

*Then, for all $h \geq 1$,*

$$\deg_j [\varepsilon^h] \log F_j \leq h + 1, \qquad [j^{h+1}\varepsilon^h] \log F_j = \frac{r^{-h} - 2}{h(h+1)}. \tag{5}$$

*All identities are coefficientwise formal identities, with the $c_z$ independent of $j$ and $\varepsilon$.*

*Proof.* Write $f(x) = \sum_{s \geq 1} f_s x^s$, where $f_1 = r$. Expansion of the exponential gives

$$[\varepsilon^h] A_j = \sum_{\sum_{s \geq 2} (s-1) m_s = h} (j)_{\underline{\sum s m_s}} \prod_{s \geq 2} \frac{(f_s / r^s)^{m_s}}{m_s!}. \tag{6}$$

This is a polynomial of degree at most $2h$.

For the stronger logarithmic bound, view $A_j$ as the partition function of disjoint subsets $B \subseteq [j]$, $|B| = s \geq 2$, with weight $s! f_s r^{-s} \varepsilon^{s-1}$. Its logarithm is the sum of connected clusters of intersecting subsets. Explicitly, the weight of an ordered tuple $B_1, \dots, B_m$ in the logarithm is its product weight divided by $m!$, multiplied by

$$\sum_{H \text{ connected on } [m]} \prod_{\{a,b\} \in E(H)} (-\mathbf{1}_{B_a \cap B_b \neq \varnothing}).$$

This identity follows by expanding the incompatibility product over all graphs and decomposing each graph into connected components; repeated subsets in the tuples are allowed. Every weight has positive $\varepsilon$-order. A connected term of order $h = \sum_a (|B_a| - 1)$ uses at most $h + 1$ labels, by adding subsets along a spanning tree of its intersection graph. Grouping terms by their union size therefore expresses

$$L_j := \log A_j = \sum_{h \geq 1} \ell_h(j) \varepsilon^h, \qquad \deg \ell_h \leq h + 1.$$

This is the degree bound already proved in [2, Theorem 3.1].

Put $B(t) = \sum_{h \geq 1} [j^{h+1}] \ell_h(j)\, t^{h+1}$. Differentiating $e^{nf(x)}$ gives, after normalization,

$$j\varepsilon = \sum_{s \geq 1} \frac{s f_s}{r^s} \varepsilon^s (j)_{\underline{s}} \exp(L_{j-s} - L_j). \tag{7}$$

The identity first holds for nonnegative integer $j$. At each $\varepsilon$-order it is polynomial in $j$, so it holds formally in $j$. Substitute $j = t/\varepsilon$. Fixed shifts lower polynomial degree: $\deg(\ell_h(j-s) - \ell_h(j)) \leq h$. Hence the shifted logarithm belongs to $\mathbb{Q}(r)[[t, \varepsilon]]$, and its $\varepsilon^0$ coefficient is $-sB'(t)$. Also $\varepsilon^s (j)_{\underline{s}} = \prod_{a=0}^{s-1} (t - a\varepsilon)$ has total degree $s$. Thus taking the $\varepsilon^0$ coefficient in (7) is legitimate and yields

$$t = x f'(x), \qquad x = \frac{t}{r} e^{-B'(t)}.$$

From (1)–(2),

$$x f'(x) = 1 - T_r(-x), \qquad (1 - r^2 y) T_r(y)^2 + (r - 2) T_r(y) - (r - 1) = 0.$$

Putting $T_r(-x) = 1 - t$ therefore gives

$$x = \frac{t(r - t)}{r^2 (1 - t)^2}, \qquad B'(t) = \log \frac{t}{rx} = 2 \log(1 - t) - \log(1 - t/r).$$

The coefficient in (5) follows for $A_j$.

Finally, series $\sum_h p_h(j) \varepsilon^h$ with $\deg p_h \leq h$ form a ring closed under formal exponentials with zero constant term and logarithms with constant term one. Each $L_{j-z} - L_j$ is in this ring. Therefore

$$F_j / A_j = 1 + \sum_{z \geq 4} \frac{c_z}{r^z} \varepsilon^z (j)_{\underline{z}} \exp(L_{j-z} - L_j)$$

is in the ring as well, and $[\varepsilon^h] \log(F_j / A_j)$ has degree at most $h$. This proves both assertions for $F_j$. □

## A finite polynomial with positive leading term

Fix $J = i + k$ and set $B_j = j! m_j / (nr)^j$ for $0 \leq j \leq J$. Treating $c_4, \ldots, c_J$ as indeterminates, (3) identifies $B_j$ with $F_j$. For integer $j$, (6) is the exact polynomial $A_j$, and terms with $z > j$ vanish. Thus every $B_j$ is a finite polynomial in $\varepsilon$ and the $c_z$, with constant term one.

Let $s_l = (-1)^{k-l}$ and define finite polynomials

$$P_+ = \prod_{s_l = 1} B_{i+l}^{\binom{k}{l}}, \qquad P_- = \prod_{s_l = -1} B_{i+l}^{\binom{k}{l}}, \qquad D = P_- - P_+.$$

The $k$th difference kills polynomials of degree less than $k$ and sends $j^k$ to $k!$. Lemma 2 gives, in the formal polynomial coefficient ring,

$$\log P_+ - \log P_- = -a_{r,k}\varepsilon^{k-1} + O(\varepsilon^k), \qquad a_{r,k} = (k-2)!\big(2 - r^{-(k-1)}\big) > 0. \tag{8}$$

Since $P_-(0) = 1$, exponentiation yields the exact finite expansion

$$D = a_{r,k}\varepsilon^{k-1} + \sum_{h=k}^{H} R_h(c_4, \dots, c_J)\varepsilon^h \tag{9}$$

for some finite $H$ and fixed polynomials $R_h$. The formal calculation determines the initial coefficients of a finite polynomial; it makes no convergence claim about a random logarithm.

Now evaluate the $c_z$ on the random graph. Each $R_h$ is a fixed polynomial in finitely many $\eta_s$, by (3), so (4) bounds its second moment. Equation (9) and the triangle inequality in $L^2$ give

$$\big\|n^{k-1}D - a_{r,k}\big\|_2 \le \sum_{h=k}^{H} n^{k-1-h}\|R_h\|_2 = O(n^{-1}).$$

Consequently,

$$\Pr\{D \le 0\} \le \frac{\mathbb{E}|n^{k-1}D - a_{r,k}|^2}{a_{r,k}^2} = O(n^{-2}). \tag{10}$$

A regular bipartite graph has a perfect matching by Hall's theorem: for each set of left vertices, counting its incident edges gives $r|S| \le r|N(S)|$. Thus $m_j > 0$ for $0 \le j \le n$. For $n \ge J$ all the $B_j, P_+, P_-$ used above are positive. Since $k \ge 2$, the term $j\log(nr)$ disappears under $\Delta^k$, and

$$\Delta^k u(i) = \log P_- - \log P_+.$$

Its sign equals that of $D$. The theorem follows from (10). □

### Attribution, verification, and scope

The matching expansion and random-graph moment bounds are established inputs, credited above. The leading coefficient is Pernici's formula; the all-order proof is reproduced from our MathDB 349137 result. The finite-product probability method follows Federbush. The new conclusion here is its unrestricted application to every fixed $r, i, k$, with the explicit failure bound. This does not establish simultaneous positivity for indices growing with $n$.

The exact verifier checks the source coefficients, logarithmic degree and difference cancellations, finite-product leading terms, and matching counts obtained independently from graph enumeration. All five pages of the current primary and all 18 pages of its companion proof paper were read. Bounded checks found no target journal final or earlier external full claim. Review is by the authoring agent, without independent peer review or proof-assistant certification.

## Tasks 2–5. Proposed follow-up directions

**Task 2.** Determine explicit constants in the failure bound in terms of $r, i, k$, using contributor-count moments and the finite polynomial in (9).

**Task 3.** Use higher bounded moments to sharpen the fixed-parameter failure probability, and identify the first nontrivial fluctuation of the scaled polynomial $n^{k-1}D$.

**Task 4.** Obtain estimates uniform in increasing $i$ and $k$, to approach the separate strong virial-positivity conjecture.

**Task 5.** Apply the same all-order coefficient calculation to weak graph positivity and to the factorial-modified matching quantity discussed at the end of [1].

## References


[1] P. Federbush, *Random Regular Bipartite Graphs Satisfy Weak Virial Positivity, for a Large Range of the Parameters*, arXiv:2107.05110v2, 2022.

[2] P. Federbush, *A PROOF of Weak Graph Positivity, for a Large Range of the Parameters*, arXiv:2012.10927v8, 2022.

[3] I. M. Wanless, *Counting matchings and tree-like walks in regular graphs*, Combin. Probab. Comput. **19** (2010), 463.

[4] M. Pernici, $1/n$ *expansion for the number of matchings on regular graphs and monomer-dimer entropy*, J. Stat. Phys. **168** (2017), 666.